

SOUND MAKERS

PCs no longer play little whooshes and tinkles. New-age entertainment applications such as DVDs and next-generation games provide a total surround sound experience... if your soundcard supports it

The thought that a bland, beige box that at best beeped an error at you through an internal speaker, would one day evolve into every audiophile's dream sound machine was unimaginable. Yet within a span of 10 years, the PC has gone from beeps and now proudly offers Dolby Digital surround sound in all its 5.1-channel glory. This mini-revolution has been headed by the unassuming soundcard that can be found nesting inside most computer systems.

A few years back, a musician would have been forced to buy a sound system that ran into lakhs of rupees. Now, thanks to the evolution of the soundcard and the introduction of features such as wavetable synthesis, 24-bit recording depth at 96 KHz, digital input and output connectors, a computer and a good soundcard will let you create professional music at an affordable price. The computer has also become the primary source of entertainment, replacing both the TV and the silver screen. Armed with the right soundcard, it can deliver theatre-quality movies in the comfort and convenience of your home. And if you enjoy playing computer games, the experience has never been better.

We approached this test with the intention of suggesting the best sound solution to suit your needs. We tested six soundcards that were available in the market. Also, as newer motherboards now come with in-built sound support that offers above average audio quality, we included two motherboards with integrated sound chipsets in our tests.

Digit Test Process

The test system consisted of an Intel Pentium 4 2.53 GHz processor on an Intel 845GBV motherboard, along with 256 MB of DDR SDRAM and an 8 GB (5,400-rpm) hard disk. All sound tests were conducted under Windows XP Professional.

As the speakers can affect the sound quality, we chose the most appropriate speakers for each card. Thus for those that supported 2.1-channel output we used Labtec Pulse 424 2.1 speakers. For both 4.1 channel and 5.1 channel soundcards Creative DTT 2500 Digital, (5.1 speakers) was used. By choosing the best speaker set for each card, we hoped to eliminate the speakers as a potential bottleneck.

Digit Test Methodology

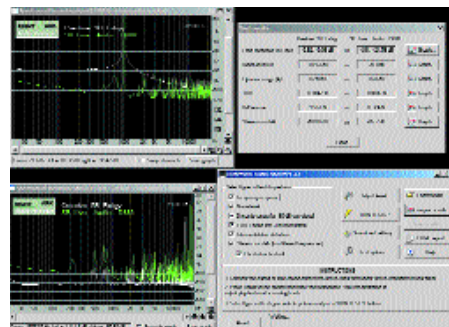
We evaluated each soundcard both in terms of what it can theoretically deliver and what it actually offers. We evaluated the different soundcards on performance (40 per cent weightage), features that they sport (30 per cent weightage), warranty and support (10 per cent weightage) and value for money (20 per cent weightage).

Features: Certain features should be looked for before purchasing a soundcard, such as the number of channels it supports, whether it is compatible with the major sound APIs and the build quality of both the card and the cables that accompany it. A card was awarded points based on these criteria.

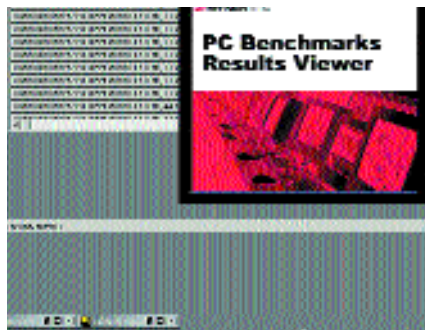
Performance: We defined the performance of a soundcard based on both subjective and analytical tests. The subjective tests evaluated the card for the music quality while playing audio CDs and the sound quality in games. Even the best soundcards stress the CPU to some extent. We used both a game and a benchmark tool to determine the impact that a soundcard might have on CPU utilisation. Finally, RightMark Audio Analyzer (RMAA) was used to determine the technical performance of a soundcard.

a) Sound quality: This was a subjective test to ascertain the performance of a card while handling both music and games. We played four different music tracks—two Hindi and two English. This helped us access the ability of a soundcard to reproduce different musical tones and notes. To determine the sound quality that a soundcard delivers under a game, we chose *Quake III Arena* (map DM9). We ran the game with High Quality sound enabled and at 1024x768 with 32-bit colour and texture depth.

b) CPU utilisation: We used *Quake III Arena* to ascertain the stress a soundcard puts on the CPU while playing back High Quality audio. We used an in-game pre-recorded demo (Demo Four) and noted the fps count under Normal and High Quality sound settings. The screen resolution was kept at 1024x768x32. We also used 3D Audio WinBench, which is an effective test that gives



The RightMark Audio Analyzer was used to determine the technical performance of a card



3D Audio WinBench was used to measure the CPU utilisation of each soundcard